

1. PROJECT OVERVIEW

- **Project Title:** FitBuddy - AI Workout Generator

2. SYSTEM COMMUNICATION PROTOCOLS:

In the context of the FitBuddy architecture, "Communication" refers to the highly synchronized data exchange between the local client, the web server, the database, and the external AI cloud infrastructure.

3. INTERNAL DATA EXCHANGE

- **Client-Server Communication:** The Jinja2 frontend communicates with the FastAPI backend by transmitting user physiological metrics (Age, Weight, Goal, Intensity) via secure HTTP POST requests.
- **Server-Database Communication:** The backend communicates with the local SQLite database utilizing SQLAlchemy ORM. It executes synchronous insert operations to persistently store user profiles and AI-generated workout logs.

4. EXTERNAL API DIALOGUE

- **Generative AI Orchestration:** The backend acts as an intermediary, translating the user's HTML form data into a structured natural language prompt. This prompt is transmitted securely over HTTPS to the Google Gemini API.
- **Fault-Tolerant Feedback Loop:** A critical communication protocol involves the system's timeout handling. If the external Gemini API fails to communicate back within the designated 10-second window (resulting in 429 or 504 errors), the Python backend intercepts the dropped connection and communicates a graceful fallback error message to the frontend UI, preventing server collapse.